Chapter 10. Linear Regression

Least squares, permutation and bootstrap inference, robust regression and M-estimation

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10.1 Simple linear regression

Model $y_i = \alpha + \beta x_i + \epsilon_i$ with uncorrelated errors of mean 0 and variance σ^2; inference is conditional on the x_i. Least squares gives $\hat\beta = S_{xy}/S_{xx}$, $\hat\alpha = \bar y - \hat\beta\bar x$ and $s^2 = \sum e_i^2/(n-2)$.

Proposition 10.1

$E\hat\beta = \beta$ and $\operatorname{Var}\hat\beta = \sigma^2/S_{xx}$. The residuals satisfy $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ with leverage $h_i = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}}$, so the adjusted residuals $r_i = e_i/\sqrt{1-h_i}$ have variance σ^2.

Proof. $\hat\beta = \sum w_i y_i$ with $w_i = (x_i - \bar x)/S_{xx}$, $\sum w_i = 0$, $\sum w_i x_i = 1$, $\sum w_i^2 = 1/S_{xx}$. With the hat matrix H, $\mathbf{e} = (I-H)\boldsymbol\epsilon$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)$, whose diagonal entries are $\sigma^2(1-h_i)$. ■

Under normal errors, $(\hat\beta - \beta)/\widehat{\mathrm{se}}(\hat\beta) \sim t_{n-2}$ with $\widehat{\mathrm{se}}(\hat\beta) = s/\sqrt{S_{xx}}$. Without normality we use resampling.

Permutation test of $H_0 : \beta = 0$. Under H_0, $y_i = \alpha + \epsilon_i$ are exchangeable, so shuffle the y's against the fixed x's. The statistics $\hat\beta$, $\sum x_i y_i$ and $t = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$ give the same test: with S_{xx} and S_{yy} fixed, $t = \hat\beta\sqrt{(n-2)S_{xx}}/\sqrt{S_{yy} - \hat\beta^2 S_{xx}}$ is increasing in $\hat\beta$. This is the same test as the permutation test of correlation (Section 5.1).

Bootstrap intervals. Use the case bootstrap (random x, robust to model failure) or the residual bootstrap $y_i^* = \hat\alpha + \hat\beta x_i + r_i^*$ (fixed design, model assumed). Because $\widehat{\mathrm{se}}(\hat\beta)$ has a formula, the bootstrap-t interval

$$\beta \in \big(\hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(1-\tfrac{\alpha}{2}),\ \hat\beta - \widehat{\mathrm{se}}(\hat\beta)Q_{t^*}(\tfrac{\alpha}{2})\big), \qquad t^* = \frac{\hat\beta^* - \hat\beta}{\widehat{\mathrm{se}}(\hat\beta^*)},$$

is cheap and second-order accurate. For tests of $\beta = 0$:

- null resampling: resample x and y *independently* (case form), or $y^* = \bar y + r_0^*$ with $r_{0,i} = (y_i - \bar y)/\sqrt{1-1/n}$ (residual form);
- pivot resampling: use t^* above and compare with $t_{\mathrm{obs}} = \hat\beta/\widehat{\mathrm{se}}(\hat\beta)$, which inverts the bootstrap-t interval.

10.2 Multiple regression

$\mathbf{y} = X\boldsymbol\beta + \boldsymbol\epsilon$ with X of size $n \times (p+1)$: $\hat{\boldsymbol\beta} = (X'X)^{-1}X'\mathbf{y}$, $E\hat{\boldsymbol\beta} = \boldsymbol\beta$, $\operatorname{Var}\hat{\boldsymbol\beta} = \sigma^2(X'X)^{-1}$, $s^2 = \mathbf{e}'\mathbf{e}/(n-p-1)$ (unbiased since $E\,\boldsymbol\epsilon'(I-H)\boldsymbol\epsilon = \sigma^2\operatorname{tr}(I-H)$). Each β_j is a *partial* effect: the effect of x_j adjusted for the other regressors.

Comparing nested models. Split $X = [X_0, X_1]$ ($k+1$ and $q = p-k$ columns) and test $H_0 : \boldsymbol\beta_1 = 0$. With hat matrices H and H_0,

$$F = \frac{\mathbf{y}'(H - H_0)\mathbf{y}/q}{\mathbf{y}'(I-H)\mathbf{y}/(n-p-1)} = \frac{\hat{\boldsymbol\beta}_1'\{X_1'(I-H_0)X_1\}\hat{\boldsymbol\beta}_1}{q\,s^2}. \tag{10.1}$$

Proposition 10.2

Under H_0 with normal errors, $F \sim F_{q,n-p-1}$. More generally $F_{\boldsymbol{\beta}_1}$, defined by replacing $\hat{\boldsymbol{\beta}}_1$ with $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1$ in (10.1), has this distribution for every $\boldsymbol{\beta}_1$, so it is a pivot.

Proof. $H - H_0$ is a projection of rank q orthogonal to $I - H$, and both annihilate X_0. Under H_0, the numerator and denominator quadratic forms reduce to $\boldsymbol{\epsilon}'(H - H_0)\boldsymbol{\epsilon}$ and $\boldsymbol{\epsilon}'(I - H)\boldsymbol{\epsilon}$, independent $\sigma^2\chi^2_q$ and $\sigma^2\chi^2_{n-p-1}$ (Cochran). The second form of (10.1) is Frisch–Waugh–Lovell: $\hat{\boldsymbol{\beta}}_1$ is the regression of $(I - H_0)\mathbf{y}$ on $\tilde{X}_1 = (I - H_0)X_1$, and $H - H_0$ projects onto the column space of $\tilde{X}_1$. For general $\boldsymbol{\beta}_1$, apply the same argument to $\mathbf{y} - X_1\boldsymbol{\beta}_1$. ∎

For $q = 1$, $F = t_j^2$, so the F-test of one coefficient is the two-sided t-test. A one-sided alternative $\beta_j > 0$ needs t_j itself, and the permutation and bootstrap schemes below apply to t_j in the same way. For bootstrap tests in multiple regression, null resampling uses either the case form (draw the $\mathbf{x}$-rows and the y's independently, which is appropriate when all slopes are tested) or residuals of the reduced model with replacement. Pivot resampling uses the case or full-model residual bootstrap with $F_{\hat{\boldsymbol{\beta}}_1}$ or $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$, and the p-value is $\#\{F^{*b} \geq F_{\mathrm{obs}}\}/(B+1)$ (with the observed value counted).

Permutation tests for $H_0 : \boldsymbol{\beta}_1 = 0$. Under H_0 the errors are exchangeable but the y_i are not (their means depend on X_0), so no permutation test is exact. Three approximate schemes:

1. *Shuffle y:* effectively tests $\beta_1 = \cdots = \beta_p = 0$; it can be anticonservative and is not recommended.
2. *Freedman–Lane:* fit the reduced model, shuffle its residuals e_0, set $y^* = \hat{y}_0 + e_0^*$, and recompute F. This is generally the most accurate.
3. *ter Braak:* shuffle the full-model residuals, set $y^* = \hat{y} + e^*$, and recompute the pivot $F_{\hat{\boldsymbol{\beta}}_1}$.

All three are asymptotically equivalent. Replacing "shuffle" with "resample with replacement" in schemes 2 and 3 gives the residual bootstrap tests under the null and with the pivot, respectively. Bootstrap-t intervals for each β_j use $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\mathrm{se}}(\hat{\beta}_j^*)$ with $\widehat{\mathrm{se}}^2(\hat{\beta}_j) = s^2(X'X)^{-1}_{jj}$.

10.3 Robust regression

Least squares is the maximum likelihood estimator under normal errors. It loses efficiency under heavy-tailed errors and can be ruined by a single outlier.

Definition (Breakdown point)

The smallest fraction of observations that, sent to infinity, can send the estimator to infinity. Roughly: the mean has $1/n \to 0$, the α-trimmed mean α, and the median $1/2$.

Trimmed and Winsorized means. The α-trimmed mean drops the lowest and highest α fractions of the data and averages the rest. The α-Winsorized mean instead replaces them by the α and $1 - \alpha$ sample quantiles and then averages all n values. Both have breakdown point α. The Hodges–Lehmann estimator has breakdown point $1 - 1/\sqrt{2} \approx 29\%$: sending a fraction ε of points to infinity drags a fraction $1 - (1 - \varepsilon)^2$ of the Walsh averages with them, which exceeds $\frac{1}{2}$ when $\varepsilon > 1 - 1/\sqrt{2}$.

Example 10.1

For $-2, -1, -1, 0, 0, 1, 1, 2, 3, 50$: mean 5.3; 10% trimmed mean 0.625 (drop -2 and 50); 10% Winsorized mean 0.7 (replace them by -1 and 3); median 0.5. Replacing 50 by 10^{100} changes only the mean.

Trimming and Winsorizing treat both tails alike, even when the contamination is one-sided. M-estimators instead down-weight observations according to their own residuals.

10.3.1 M-estimators

For location, the least squares estimator $\bar{x} = \arg\min_\mu \sum (x_i - \mu)^2$ is generalized to

$$\hat{\mu}^\Psi = \arg\min_\mu \sum_i \Psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) \iff \sum_i \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi = \Psi',$$

with a robust scale $\hat{\tau} = 1.4826$ MAD (consistent for σ under normality, since $\Phi^{-1}(3/4) = 0.6745$).

- $\psi(x) = x$: the mean. $\psi(x) = \operatorname{sgn}(x)$: the median.
- **Huber:** $\psi(x) = \max(-k, \min(k, x))$, quadratic Ψ in the center and linear in the tails. With $k \approx 1.35$ it has about 95% efficiency at the normal.
- **Tukey biweight:** $\psi(x) = x(1 - (x/c)^2)^2 I(|x| \le c)$, redescending, so gross outliers get zero weight.

A bounded ψ with a 50% breakdown scale gives breakdown point 50%. ("M" stands for maximum-likelihood type: with $\Psi(x) = x^2/2$ and $\hat{\tau} = \sigma$, $\hat{\mu}^\Psi$ is the normal maximum likelihood estimator.) The equation has no closed form and is solved numerically, e.g. by iteratively reweighted least squares with weights $\psi(u_i)/u_i$.

Inference on a location with outliers. With a contaminated sample, the t-test and t-interval are driven by the outliers: the mean and s both inflate. Robust alternatives:

- the sign test and order-statistic interval (Chapter 2), valid for the median under any continuous F;
- the signed-rank test and Walsh-average interval, valid if the uncontaminated distribution is symmetric;
- an M-estimator with a bootstrap standard error and bias, giving a normal-approximation bootstrap interval, or better a percentile or BCa interval. Its asymptotic standard error requires the influence function, so the bootstrap is the practical route.

The lower the breakdown point of the underlying estimator, the wider the resulting interval tends to be under contamination. In regression, $\hat{\boldsymbol{\beta}}^\Psi = \arg\min \sum \Psi((y_i - \mathbf{x}_i'\boldsymbol{\beta})/\hat{\tau})$. The estimating equations $\sum \psi(\cdot)\,\mathbf{x}_i = 0$ show that Huber's estimator is robust to outlying y but not to outlying x (high leverage), because the factor x_{ij} is unbounded. **Yohai's MM-estimator** uses a redescending ψ (biweight, $c = 4.685$ for 95% efficiency) started from a high-breakdown S-estimate of scale, achieving a 50% breakdown point together with high efficiency. **Least trimmed squares** minimizes the sum of the h smallest squared residuals; $h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ gives maximal breakdown, and $h = \lfloor 0.9n \rfloor$ a 10% trimming. **Least median of squares** minimizes the median squared residual. Both are very robust but inefficient, and are mainly used as starting values.

10.3.2 The Theil–Sen estimator

For $y_i = \alpha + \beta x_i + \epsilon_i$, form the pairwise slopes $\beta_{ij} = (y_j - y_i)/(x_j - x_i)$ for $x_i \neq x_j$ and set

$$\hat{\beta}^{TS} = \text{median}_{i<j}\, \beta_{ij}, \qquad \hat{\alpha}^{TS} = \text{median}_i(y_i - \hat{\beta}^{TS} x_i).$$

Proposition 10.3 (Link to Kendall's tau)

Let $\tau_n(\beta_0)$ be Kendall's statistic between x_i and $y_i - \beta_0 x_i$. Then $\binom{n}{2}\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\}$. Hence the test of $H_0 : \beta = \beta_0$ by Kendall's tau, and the interval $(\beta_{(l)}, \beta_{(u)})$ with $l = \binom{n}{2}\frac{Q_K(\alpha/2)+1}{2}$, $u = \binom{n}{2}\frac{Q_K(1-\alpha/2)+1}{2}$ (Q_K the null quantile of τ_n), are exact. $\hat{\beta}^{TS}$ is the corresponding point estimate.

Proof. $\text{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] = \text{sgn}[(x_i - x_j)^2(\beta_{ij} - \beta_0)] = \text{sgn}(\beta_{ij} - \beta_0)$. At the true β, $y_i - \beta x_i = \alpha + \epsilon_i$ is independent of x_i, so $\tau_n(\beta)$ has the null law K_n. The acceptance region $Q_K(\frac{\alpha}{2}) \le \tau_n(\beta_0) \le Q_K(1 - \frac{\alpha}{2})$ is a range of $\#\{\beta_{ij} > \beta_0\}$, which is a step function of β_0 dropping by one at each ordered slope. So the region is an interval of ordered slopes with the stated indices. ■

$\hat{\beta}^{TS}$ has breakdown point $1 - 1/\sqrt{2} \approx 29\%$ and about 96% ARE relative to least squares under normal errors.

Example 10.2

Points $(1,1), (2,2), (3,3.5), (4,4), (5,20)$, where the last point is an outlier. The ten ordered pairwise slopes are $0.5, 1, 1, 1, 1.25, 1.5, 4.75, 6, 8.25, 16$, so $\hat{\beta}^{TS} = 1.375$ and $\hat{\alpha}^{TS} = -0.625$. Least squares gives slope 4.0, pulled up by a single point.

In multiple regression Theil–Sen has no unique definition. One version uses **backfitting**:

1. Start with single-variable Theil–Sen slopes $\hat{\beta}_j^{(0)}$ and set $\hat{\beta}_0^{(0)} = \text{median}_i(y_i - \sum_j \hat{\beta}_j^{(0)} x_{ij})$.
2. For $k = 1, 2, \ldots$ and each j, compute the Theil–Sen slope of the partial responses $y_i - \sum_{l \neq j} \hat{\beta}_l^{(k-1)} x_{il}$ on x_{ij}, then update the intercept as a median.
3. Iterate until the coefficients stabilize.

Inference with robust regression estimators. M- and MM-estimates come with approximate standard errors from the final weighted least squares step. These are reasonable for large n, and the bootstrap standard errors are usually somewhat larger and more reliable. With an approximate standard error one can form a bootstrap-t interval $(\hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(1 - \frac{\alpha}{2}), \hat{\beta}_j - \widehat{\text{se}}\, Q_{t^*}(\frac{\alpha}{2}))$ from case resampling, or permutation tests of $t_j = \hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$. Shuffling y alone tests the joint null that all slopes vanish; for a single coefficient use a residual-shuffling scheme as in Section 10.2. Bootstrap samples contain many repeated points, which can upset estimators based on pairwise slopes (ties in x); permutation tests avoid this.